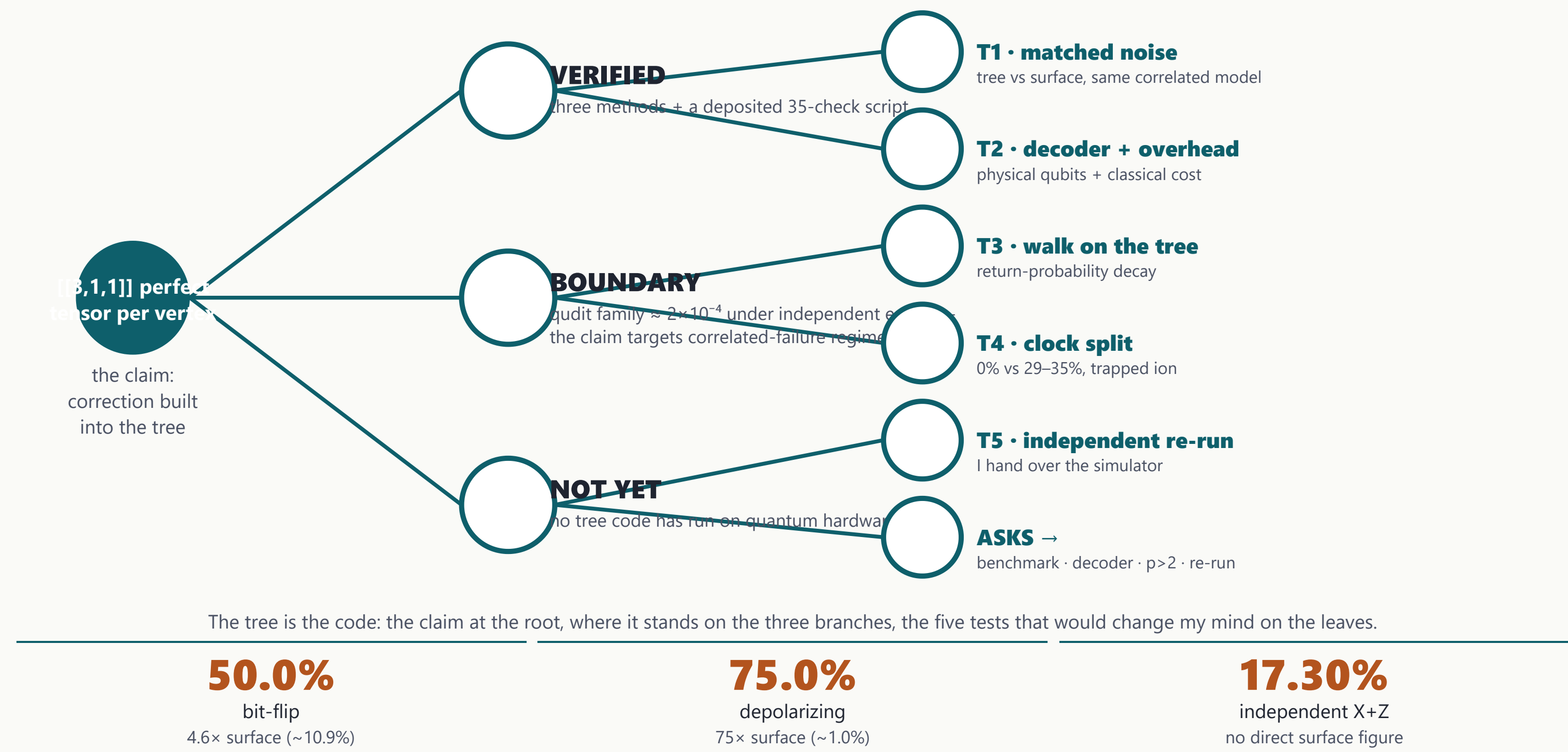


# The Bruhat–Tits Tree Code — Status and Test Plan

Rowan Brad Quni-Gudzinis · QNFO · CWI Summer School on Quantum Algorithms & QEC · Wed 26 Aug 2026, 16:30–18:00



## The tests

## T1 • Matched-noise comparison

Tree vs surface under the *same* correlated noise model. Classical, days-weeks.

**Would change my mind:** the advantage falls to parity.

## T2 · Decoder + full overhead

Physical qubits plus classical decoding cost. Classical.

**Would change my mind:** the advantage disappears once decoding cost is counted.

### T3 · Random walk on a p-regular tree

Return-probability decay. Ion / analog simulator, ~8 weeks.

**Would change my mind:** no change in the decay exponent where predicted.

#### T4 • Clock-rest split

Diagonal 0% vs non-diagonal 29–35% violation rate. Trapped ion, ~8 weeks.

**Would change my mind:** the measured split falls outside that range.

### T5 · Independent re-run

An outside group repeats T1 with my simulator and the protocol.

**Would change my mind:** nothing to predict — this one is about reproducibility.

## For this room

1. A correlated-noise benchmark — model and data — I can run T1 against.
2. A decoder reality check: does a known decoder family already cover tree-structured codes, and how does it scale?
3. Pointers on perfect tensors for  $p > 2$  (the binary  $[[3,1,1]]$  case is settled).
4. An independent re-run of T1.

**Verified:** thresholds computed three independent ways — analytical recursion, exact stabilizer simulation, Monte Carlo — and re-checked by a deposited 35-check script.

**Boundary:** under independent errors the qudit family sits at  $\approx 2 \times 10^{-4}$ , about  $55\times$  below surface codes; the advantage claim targets correlated-failure regimes — which is exactly what T1 probes.



